F1D

CEO Communication Clarity & Market Outcomes

112,968	17	4	23
EARNINGS CALLS	YEARS (2002-2018)	PIPELINE STEPS	SCRIPTS

Pipeline Overview

Raw Data → **Step 1: Sample** → Step 2: Text → Step 3: Financial → Step 4: Econometric

STEP 1: SAMPLE MANIFEST

- > Clean metadata & filter events
- > Link to Compustat/CRSP
- > Build CEO tenure map
- > Assemble final sample

RESEARCH QUESTIONS

- > Is CEO clarity a persistent trait?
- > Does unclear speech affect liquidity?
- > Do vague CEOs become targets?

1.1 CleanMetadata.py

> *Loads raw Unified-info, deduplicates records, resolves collisions, filters for earnings calls (event_type='1') in date range 2002-2018.*

INPUT

Unified-info.parquet

METRIC	VALUE
Total rows	**465,434**
Columns	**30**
Unique file_name	**428,330**

OUTPUT

metadata_cleaned.parquet

METRIC	VALUE
Total rows	**297,547**
Retention rate	**63.9%**
Unique companies	**~8,500**

PROCESS

- > Deduplicate exact rows
- > Resolve file_name collisions (keep earliest)
- > Filter event_type = '1' (earnings)
- > Filter date range 2002-2018

KEY INSIGHTS

- > **36%** filtered (non-earnings, duplicates)
- > Date coverage: 2002-01-01 to 2018-12-31
- > Ready for entity linking

1.2 LinkEntities.py

> *Links cleaned metadata to Compustat/CRSP via GVKEY matching using PERMNO, CUSIP, and fuzzy company name matching. Assigns Fama-French industry codes.*

INPUTS

`metadata_cleaned.parquet` + `CRSPCompustat_CCM.parquet`

METRIC	VALUE
Cleaned metadata	**297,547**
CCM records	**32,421**
Unique GVKEYs in CCM	**27,899**

OUTPUT

`metadata_linked.parquet`

METRIC	VALUE
Total rows	**212,389**
Retention rate	**71.4%**
GVKEY match rate	**100%**

PROCESS

- > Match by PERMNO (primary)
- > Match by CUSIP (fallback)
- > Fuzzy match company names (last resort)
- > Assign FF12 and FF48 industry codes

KEY INSIGHTS

- > **28.6%** unmatched (no Compustat link)
- > All retained calls have valid GVKEY
- > FF12 codes enable industry filtering
- > Ready for CEO matching

1.3 BuildTenureMap.py

> *Builds CEO tenure map from Execucomp data. Identifies CEO periods per GVKEY, creating monthly-level records for tenure matching.*

INPUT

comp_execucomp.parquet

METRIC	VALUE
Total exec records	**370,545**
Unique executives	**~45,000**
CEO records (ceoann='CEO')	**~28,000**

OUTPUT

tenure_monthly.parquet

METRIC	VALUE
Total tenure-months	**997,699**
Unique CEOs	**~6,800**
Unique firms (GVKEYs)	**~3,200**

PROCESS

- > Filter to CEO flag (ceoann='CEO')
- > Expand annual records to monthly
- > Identify tenure start/end dates
- > Track predecessor CEO (prev_ceo_id)

KEY INSIGHTS

- > **3x expansion** (annual → monthly)
- > Average tenure: ~12 years per CEO
- > Enables call-to-CEO time matching
- > Critical for fixed effects identification

1.4 AssembleManifest.py

> *Merges linked metadata with CEO tenure map. Filters to CEOs with 5+ calls to ensure sufficient observations for fixed effects estimation.*

INPUTS

metadata_linked.parquet + tenure_monthly.parquet

METRIC	VALUE
Linked metadata	**212,389**
Tenure months	**997,699**

PROCESS

> Match calls to CEO by (GVKEY, date)

> Assign ceo_id to each call

> Count calls per CEO

> Filter: keep CEOs with ≥5 calls

OUTPUT

master_sample_manifest.parquet

METRIC	VALUE
Final sample	**112,968**
Retention rate	**53.2%**
Unique CEOs	**~4,200**
Unique firms	**~2,800**

KEY INSIGHTS

> **46.8%** filtered (CEO matching + min calls)

> Avg **27 calls** per CEO

> Final sample ready for textual analysis

> Sufficient variation for fixed effects

2.1 TokenizeAndCount.py

> *Tokenizes speaker text from earnings call transcripts and counts Loughran-McDonald dictionary words. Produces linguistic counts at the speaker-turn level.*

INPUTS

speaker_data_{year}.parquet + LM_Dictionary.csv

+ master_sample_manifest.parquet

METRIC	VALUE
Speaker records (17 years)	**27,831,805**
Manifest files (filter)	**112,968**
LM tracked words	**3,859**

PROCESS

- > Filter to manifest file_names
- > Uppercase text, alpha-only tokens
- > CountVectorizer with LM vocabulary
- > Aggregate counts per category

OUTPUT

linguistic_counts_{year}.parquet

METRIC	VALUE
Output records	**9,823,323**
Retention (manifest filter)	**35.3%**
New columns added	**8**

CATEGORIES

- > **Negative**, **Positive**
- > **Uncertainty**, **Litigious**
- > Strong_Modal, Weak_Modal, Constraining
- > total_tokens (word count)

2.2 constructvariables.py

> *Aggregates speaker-level word counts to call-level linguistic percentages. Flags speakers as Manager/Analyst/CEO and computes weighted averages across samples and contexts.*

INPUTS

`linguistic_counts_{year}.parquet`

+ `master_sample_manifest.parquet`
+ `managerial_roles_extracted.txt`

METRIC	VALUE
Speaker-level records	**9,823,323**
Manifest calls	**112,968**
Manager keywords	**45**

PROCESS

> Flag speakers: is_manager, is_analyst, is_ceo

> Group by file_name (call)

> Compute weighted pct = sum(count)/sum(tokens) ×

OUTPUT

`linguistic_variables_{year}.parquet`

METRIC	VALUE
Call-level rows	**112,968**
Unique GVKEYs	**2,429**
Total columns	**110**
Linguistic variables	**105**

VARIABLE STRUCTURE

> **5 Samples:** Manager, Analyst, CEO, NonCEO_Manager, Entire

> **3 Contexts:** QA, Pres, All

> **7 Categories:** Negative, Positive, Uncertainty, etc.

> 5 × 3 × 7 = **105** linguistic variables

3.1 FirmControls.py

> *Computes firm-level control variables from Compustat, IBES, and CCCL. Matches to manifest calls via GVKEY.*

INPUTS

master_sample_manifest (2,429 GVKEYs)

+ Compustat (~25K GVKEYs)
+ IBES (21K CUSIPs)
+ CCCL (18K GVKEYs)

PROCESS

> Match Compustat quarterly to GVKEY+date
> Match IBES EPS by CUSIP+period
> Compute earnings surprise decile
> Merge CCCL shift_intensity

OUTPUT

firm_controls_{year}.parquet

METRIC	VALUE
Total rows	112,968
Unique GVKEYs	2,429 (100%)

VARIABLE	COVERAGE
Size, Lev, ROA	99.8%
BM | SurpDec	98.8% | 76.3%
shift_intensity	99.6%

VARIABLES

> **Size:** ln(atq) | **BM:** ceqq/(cshoq×prccq)
> **Lev:** ltq/atq | **ROA:** niq/atq
> **SurpDec:** Earnings surprise decile (-5 to +5)
> **shift_intensity:** CCCL (6 variants)

Computes stock returns and liquidity measures from CRSP daily data. Uses PERMNO linkage via manifest and CCM fallback.

INPUTS

master_sample_manifest (2,429 GVKEYs)

+ CRSP_DSF (96 quarterly files, 1999-2022)

+ CCM (27,899 GVKEYs → 28,503 PERMNOs)

PROCESS

> Map GVKEY → PERMNO via CCM fallback

> Compute returns: prev_call+5d to call-5d

> Compound StockRet and MarketRet (vwretd)

> Compute Amihud illiquidity: mean(|ret|/vol)

> Corwin-Schultz spread estimator

OUTPUT

market_variables_{year}.parquet

METRIC	VALUE
Total rows	112,968
Columns	11

VARIABLE	COVERAGE
StockRet	94.8%
MarketRet	95.0%
Volatility	94.8%
Amihud	98.1%
Corwin_Schultz	98.1%

VARIABLES

> **StockRet:** Compound return (%)

> **MarketRet:** VW market return (%)

> **Volatility:** Annualized std dev

3.3 EventFlags.py

> *Computes takeover event flags from SDC M&A data. Matches calls to takeover announcements within 365 days via CUSIP6.*

INPUTS

master_sample_manifest (2,429 GVKEYs)

\+ SDC M&A (sdc-ma-merged.parquet)

SDC INPUT	VALUE
Total deals	142,457
Unique target CUSIPs	132,153
Date range	1999-2025

DEAL ATTITUDE	COUNT
Friendly	136,541 (95.8%)
Unsolicited	1,564 (1.1%)
Hostile	360 (0.3%)

OUTPUT

event_flags_{year}.parquet

METRIC	VALUE
Total rows	112,968
Takeover = 1	1,132 (1.00%)
Takeover = 0	111,836

TAKEOVER TYPE	COUNT
Friendly	~85%
Uninvited (hostile+unsolicited)	~15%

VARIABLES

> **Takeover:** Binary (1 if target within 365d)

> **Takeover_Type:** "Friendly" or "Uninvited"

> **Duration:** Quarters until event (survival)

4.1 Model Specification & Variable Correlations

$$Manager_QA_Uncertainty_{it} = \alpha + \gamma_i + \delta_t + \beta_1 Pres_Unc_{it} + \beta_2 Analyst_Unc_{it} + \beta_3 Negative_{it} + \beta_4 StockRet_{it} + \beta_5 MktRet_{it} + \beta_6 EPSGrow_{it} + \beta_7 SurpDec_{it} + \varepsilon_{it}$$

VARIABLE DEFINITIONS

VARIABLE	DEFINITION
Manager_QA_Unc	% uncertainty words by manager in Q&A (DV)
γ_i (CEO FE)	CEO fixed effect → becomes ClarityCEO
δ_t (Year FE)	Year fixed effects (2002-2018)
Pres_Unc	% uncertainty words in presentation section
Analyst_Unc	% uncertainty words by analysts in Q&A
Negative	% negative words (Loughran-McDonald)
StockRet	Compound stock return (prev_call+5d to call-5d)
MktRet	Value-weighted market return (same window)
EPSGrow	Quarterly EPS growth rate
SurpDec	Earnings surprise decile (-5 to +5, from IBES)

CORRELATION MATRIX (N=57,065)

4.1 EstimateCeoClarity.py

> *Estimates CEO "Clarity" via OLS fixed effects. DV: Manager_QA_Uncertainty_pct. Retention: ~51% (Main Sample).*

DATA RETENTION CASCADE (MAIN SAMPLE)

FILTER STEP	OBS	LOST
Start (manifest)	112,968	
Main only (excl. FF12=8,11)	74,948	-38,020
Manager_QA_Uncertainty	69,910	-5,038
Analyst_QA_Uncertainty	66,834	-3,076
Financial Controls (Stock/EPS)	61,657	-5,177
SurpDec (IBES)	57,694	-3,963
CEO ≥5 calls	57,065	-629

WHY NULLS? (LINGUISTIC VARS)

> **Manager_QA:** No manager spoke in Q&A

> **Analyst_QA:** No analyst questions

> **SurpDec:** No IBES forecast coverage

OUTPUT

ceo_clarity_scores.parquet

SAMPLE	CEOS
Main (non-fin/util)	2,587
Finance (FF12=11)	569
Utility (FF12=8)	135
Total	3,291

MODEL FIT (MAIN)	VALUE
Observations	48,628
R^2	0.410
Adj. R^2	0.382

KEY OUTPUT

> **ClarityCEO:** Standardized (μ=0, σ=1)

> Higher = less uncertainty = clearer

4.1 Regression Results (Main Sample)

> *Results confirm valid CEO Clarity extraction: Strong R^2 (41%) and expected correlations with controls.*

REGRESSION ESTIMATES

DV: Manager_QA_Uncertainty_pct

VARIABLE	COEF.	T-STAT	P-VAL	SIG.
Manager_Pres_Uncertainty	0.0850	16.77	0.000	***
Analyst_QA_Uncertainty	0.0335	11.56	0.000	***
Negative_pct	0.0745	13.52	0.000	***
SurpDec (Earnings Surprise)	0.0011	2.84	0.004	***
MarketRet	-0.0006	-2.96	0.003	***
EPS_Growth	0.0008	1.95	0.052	*
StockRet	-0.0001	-1.33	0.185	

** p<0.1, ** p<0.05, *** p<0.01. HC1 Robust Errors.*

MODEL DIAGNOSTICS

METRIC	VALUE
Observations	57,065
R-squared	0.410
Adj. R-squared	0.382
F-statistic	18.82
CEO Fixed Effects	Included
Year Fixed Effects	Included

INTERPRETATION

> High R^2 (41%): The model explains a significant portion of variation in manager uncertainty.

> Controls work: Expected positive correlation with presentation uncertainty and negative sentiment.

> Residuals: The remaining unexplained variance (after removing controls + FE) is the "pure" CEO Clarity trait.

> *Full regression specifications showing the determinants of Manager QA Uncertainty across samples.*
> *Includes CEO Fixed Effects (omitted for brevity) and linguistic/financial controls.*

ACADEMIC REGRESSION TABLE (DEP VAR: MANAGER_QA_UNCERTAINTY_PCT)

VARIABLE	MAIN	FINANCE	UTILITY
Intercept	1.005*** (0.000)	0.826*** (0.000)	0.593*** (0.000)
Manager_Pres_Uncertainty	0.085*** (0.000)	0.085*** (0.000)	0.144*** (0.000)
Analyst_QA_Uncertainty	0.034*** (0.000)	0.044*** (0.000)	0.054*** (0.000)
Entire_All_Negative	0.075*** (0.000)	0.061*** (0.000)	0.001 (0.968)
StockRet	-0.000 (0.185)	0.000 (0.449)	-0.001* (0.086)
MarketRet	-0.001*** (0.003)	-0.001* (0.076)	0.001 (0.171)
EPS_Growth	0.001* (0.052)	-0.001 (0.565)	-0.001 (0.704)
SurpDec	0.001*** (0.004)	-0.001 (0.392)	-0.003 (0.105)
CEO Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Observations	57,065	13,239	2,955
R-squared	0.410	0.305	0.217
Adj. R-squared	0.382	0.273	0.173

Alternative model using CEO-only speech (excludes CFO/other managers). DV: CEO_QA_Uncertainty_pct

MODEL FORMULA

$$\textbf{CEO_QA_Uncertainty}_{it} = \alpha + \gamma_i + \delta_t + \beta_1 CEO_Pres_Unc_{it} + \beta_2 Analyst_Unc_{it} + \beta_3 Negative_{it} + \beta_4 StockRet_{it} + \beta_5 MktRet_{it} + \beta_6 EPSGrow_{it} + \beta_7 SurpDec_{it} + \varepsilon_{it}$$

VARIABLE DEFINITIONS

VARIABLE	DEFINITION
CEO_QA_Uncertainty_pct (DV)	% uncertainty words by CEO only in Q&A
γ_i (ClarityCEO)	CEO fixed effect (becomes clarity score)
δ_t (Year FE)	Year fixed effects (2002-2018)
CEO_Pres_Unc	% uncertainty words by CEO in Presentation
Analyst_Unc	% uncertainty words by analysts in Q&A
Negative	% negative words (Loughran-McDonald)
StockRet/MktRet	Stock and market returns before call
EPSGrow/SurpDec	EPS growth and earnings surprise decile

DIFFERENCE FROM 4.1 (MANAGER MODEL)

> 4.1: Manager_QA_Uncertainty (CEO + CFO + Others)

> 4.1.1: CEO_QA_Uncertainty (CEO only)

This isolates the CEO's personal communication style, excluding CFO or IR statements.

CORRELATION MATRIX (N=42,062)

SAMPLE COMPARISON

4.1.1 EstimateCeoClarity_CeoSpecific.py - Data Retention

Retention cascade for CEO-only model. Main constraint: ~15k calls have no CEO Q&A speech.

DATA RETENTION CASCADE (MAIN SAMPLE)

FILTER STEP	OBS	LOST
Start (manifest)	112,968	
Main only (excl. FF12=8,11)	74,948	-38,020
CEO_QA_Uncertainty	~55,000	~-20,000
Financial Controls	~43,000	~-12,000
CEO ≥5 calls	42,062	~-1,000

WHY LOWER N THAN 4.1?

> **CEO_QA_Uncertainty:** Many calls have CFO/IR leading Q&A, not CEO.

> **Stricter Filter:** Only counts CEO-attributed words.

OUTPUT

ceo_clarity_scores.parquet

SAMPLE	CEOS
Main (non-fin/util)	2,013
Finance (FF12=11)	381
Utility (FF12=8)	89
Total	2,483

MODEL FIT (MAIN)	VALUE
Observations	42,062
R^2	0.345
Adj. R^2	0.311

KEY OUTPUT

> **ClarityCEO:** Standardized (μ=0, σ=1)

> Higher = less uncertainty = clearer

4.1.1 Regression Results (Main Sample)

> *CEO-only model confirms validity: R^2 (34.5%) with expected control correlations.*

REGRESSION ESTIMATES

DV: CEO_QA_Uncertainty_pct

VARIABLE	COEF.	T-STAT	P-VAL	SIG.
CEO_Pres_Uncertainty	0.0816	12.50	0.000	***
Analyst_QA_Uncertainty	0.0332	8.01	0.000	***
Negative_pct	0.0647	7.86	0.000	***
MarketRet	-0.0011	-3.44	0.001	***
EPS_Growth	0.0006	0.87	0.382	
SurpDec	0.0005	0.78	0.437	
StockRet	0.00002	0.19	0.853	

** p<0.1, ** p<0.05, *** p<0.01. HC1 Robust Errors.*

MODEL DIAGNOSTICS

METRIC	VALUE
Observations	42,062
R-squared	0.345
Adj. R-squared	0.311
F-statistic	16.11
CEO Fixed Effects	Included
Year Fixed Effects	Included

INTERPRETATION

> R^2 (34.5%): Slightly lower than 4.1 (41%) due to stricter CEO-only filter.

> CEO_Pres_Unc: Strong positive effect (t=12.5) confirms presentation anchors Q&A tone.

> MarketRet: Significant negative effect—bad market days increase CEO uncertainty.

> SurpDec/EPS: Not significant in CEO-only model.

4.1.1 Regression Results: CEO-only Utility Models

Comparative regression results for CEO-only speech across Main, Finance, and Utility samples.

REGRESSION TABLE (DEP VAR: CEO_QA_UNCERTAINTY_PCT)

VARIABLE	MAIN	FINANCE	UTILITY
Intercept	0.702*** (0.000)	0.699*** (0.002)	0.208 (0.446)
CEO_Pres_Uncertainty	0.082*** (0.000)	0.080*** (0.000)	0.056 (0.343)
Analyst_QA_Uncertainty	0.033*** (0.000)	0.043*** (0.000)	0.037 (0.205)
Entire_All_Negative	0.065*** (0.000)	0.050*** (0.008)	0.014 (0.852)
StockRet	0.000 (0.853)	-0.000 (0.937)	-0.004*** (0.007)
MarketRet	-0.001*** (0.001)	-0.000 (0.786)	0.005** (0.026)
EPS_Growth	0.001 (0.382)	0.000 (0.887)	0.002 (0.700)
SurpDec	0.001 (0.437)	-0.001 (0.744)	-0.002 (0.689)
CEO Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Observations	42,062	8,225	1,716
R-squared	0.345	0.294	0.161
Adj. R-squared	0.311	0.258	0.102
F-statistic	16.11	13.05	5.33

4.1.2 EstimateCeoClarity_Extended.py - Robustness Check

Compares baseline vs extended controls to test robustness of CEO fixed effects (Clarity).

4 MODELS TESTED

MODEL	DV	CONTROLS
Manager_Baseline	Manager_QA_Uncertainty	Base (7)
Manager_Extended	Manager_QA_Uncertainty	Base + Extended (14)
CEO_Baseline	CEO_QA_Uncertainty	Base (7)
CEO_Extended	CEO_QA_Uncertainty	Base + Extended (14)

BASE CONTROLS (7)

- > Manager/CEO_Pres_Uncertainty
- > Analyst_QA_Uncertainty
- > Negative_pct
- > StockRet, MarketRet
- > EPS_Growth, SurpDec

EXTENDED CONTROLS (+7)

VARIABLE	DEFINITION
Size	Log(Total Assets)
BM	Book-to-Market Ratio
Lev	Leverage (Debt/Assets)
ROA	Return on Assets
CurrentRatio	Current Assets / Current Liabilities
RD_Intensity	R&D / Sales
Volatility	Stock Return Volatility

RESEARCH QUESTION

Do the CEO fixed effects change when controlling for additional firm characteristics?

Robust if: R^2 barely changes and extended controls are mostly insignificant.

4.1.2 Baseline vs Extended - Model Fit Comparison

> *Extended controls add minimal explanatory power (+0.2pp R^2 for Manager, +0pp for CEO).*

MANAGER MODELS (MAIN SAMPLE)

METRIC	BASELINE	EXTENDED	Δ
Observations	57,065	55,694	-1,371
N CEOs	2,587	2,537	-50
R^2	0.410	0.412	+0.2pp
Adj. R^2	0.382	0.383	+0.1pp
F-statistic	18.82	18.89	–

CEO-ONLY MODELS (MAIN SAMPLE)

METRIC	BASELINE	EXTENDED	Δ
Observations	42,062	40,978	-1,084
R^2	0.345	0.345	+0.0pp

INTERPRETATION

> R^2 unchanged: Extended controls add ~0pp explanatory power for CEO model.

> Sample loss: ~2-3% fewer obs due to missing extended control data.

> Robustness confirmed: CEO fixed effects are stable regardless of additional firm controls.

CONCLUSION

CEO Clarity is robust.

Adding Size, BM, Leverage, ROA, Current Ratio, R&D Intensity, and Volatility does not meaningfully improve model fit or alter CEO fixed effects.

4.1.2 Extended Controls Coefficients (Manager_Extended)

> *Most extended controls are not statistically significant. Only Lev, CurrentRatio, and Volatility show weak effects.*

BASE CONTROLS (EXTENDED MODEL)

VARIABLE	COEF.	T-STAT	P-VAL	SIG.
Manager_Pres_Unc	**0.0851**	**16.41**	**0.000**	***
Analyst_QA_Unc	**0.0338**	**11.54**	**0.000**	***
Negative_pct	**0.0747**	**13.11**	**0.000**	***
MarketRet	**-0.0005**	**-2.28**	**0.022**	**
SurpDec	**0.0010**	**2.54**	**0.011**	**
StockRet	**-0.0001**	**-1.49**	**0.138**	
EPS_Growth	**0.0006**	**1.31**	**0.189**	

EXTENDED CONTROLS (NEW)

VARIABLE	COEF.	T-STAT	P-VAL	SIG.
Size	**-0.0038**	**-0.93**	**0.355**	
BM	**-0.0094**	**-1.72**	**0.085**	*
Lev	**-0.0433**	**-3.48**	**0.000**	***
ROA	**0.0396**	**0.90**	**0.370**	
CurrentRatio	**0.0026**	**2.00**	**0.046**	**
RD_Intensity	**0.0593**	**0.41**	**0.685**	
Volatility	**0.0003**	**2.69**	**0.007**	***

** p<0.1, ** p<0.05, *** p<0.01. HC1 Robust.*

KEY FINDING

> **Lev:** Higher leverage → less uncertainty (-0.043***)

> **Volatility:** Minimal effect (+0.0003***)

> Other extended controls: Not significant

Full specification with extended firm controls across all samples.
Demonstrates robustness of findings to firm size, leverage, and performance metrics.

REGRESSION TABLE (DEP VAR: CEO_QA_UNCERTAINTY_PCT)

VARIABLE	MAIN	FINANCE	UTILITY
Intercept	0.726*** (0.000)	0.437* (0.091)	0.365 (0.762)
CEO_Pres_Uncertainty	0.083*** (0.000)	0.091*** (0.002)	0.055 (0.356)
Analyst_QA_Uncertainty	0.034*** (0.000)	0.039* (0.079)	0.034 (0.232)
Entire_All_Negative	0.067*** (0.000)	0.120*** (0.001)	0.009 (0.901)
Size	-0.002 (0.759)	0.063** (0.029)	-0.048 (0.709)
BM	-0.006 (0.412)	0.004 (0.935)	0.033 (0.718)
Lev	-0.041** (0.021)	-0.046 (0.646)	0.446 (0.383)
ROA	0.081 (0.193)	0.512 (0.158)	-0.522 (0.807)
CurrentRatio	0.001 (0.625)	-0.000 (0.904)	-0.012 (0.828)
RD_Intensity	0.156 (0.487)	3.796*** (0.002)	0.000 (—)
Volatility	0.000 (0.553)	-0.000 (0.543)	-0.000 (0.923)
Base Controls	Yes	Yes	Yes
CEO Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Observations	40,978	1,660	1,716
R-squared	0.345	0.324	0.161

4.1.3 EstimateCeoClarity_Regime.py - Tone at the Top Test

> *Tests if CEO's uncertain language influences other executives' speech ("Tone at the Top"). Does the CEO's communication style set a regime that affects the board?*

MODEL FORMULA

$$\textbf{NonCEO_Manager_QA_Unc}_{it} = \alpha + \gamma_i + \delta_t + \beta_1 \cdot \text{Manager_Pres_Unc} + \beta_2 \cdot \text{Analyst_QA_Unc} + \beta_3 \cdot \text{Negative} + \beta_4 \cdot \text{StockRet} + \beta_5 \cdot \text{MarketRet} + \beta_6 \cdot \text{EPS_Growth} + \beta_7 \cdot \text{SurpDec} + \varepsilon_{it}$$

VARIABLE DEFINITIONS

SYMBOL	VARIABLE	DEFINITION
DV	NonCEO_Manager_QA_Unc	% uncertain words in CFO/IR/Other Q&A
γ_i	CEO FE	CEO fixed effect (captures CEO regime)
δ_t	Year FE	Year fixed effect
β_1	Manager_Pres_Unc	Presentation uncertainty
β_2	Analyst_QA_Unc	Analyst question uncertainty
β_3	Negative_pct	Negative sentiment % in call
β_{4-7}	Firm Controls	StockRet, MarketRet, EPS_Growth, SurpDec

RESEARCH QUESTION

Does the CEO's uncertain language influence other executives?

If CEO sets a "Tone at the Top" → CEO FE should explain Non-CEO speech too.

DV COMPARISON ACROSS SCRIPTS

SCRIPT	DV	WHO'S SPEAKING
4.1	Manager_QA_Unc	CEO + CFO + IR + Others
4.1.1	CEO_QA_Unc	CEO only
4.1.3	**NonCEO_Manager_Unc**	**CFO + IR + Others (NO CEO)**

> *R^2 drops from 41.0% (CEO+Others) to 16.3% (Non-CEO only). CEO fixed effects explain 2.5x less variance in other executives' speech.*

MODEL FIT COMPARISON (MAIN SAMPLE)

METRIC	4.1 (MANAGER)	4.1.3 (NONCEO)	Δ
DV	Manager_QA_Unc	**NonCEO_Manager_Unc**	—
Observations	**57,065**	**55,985**	**-1,080**
N CEOs	**2,587**	**2,574**	**-13**
R^2	**0.410**	**0.163**	**-24.7pp**
Adj. R^2	**0.382**	**0.122**	**-26.0pp**
F-statistic	**18.82**	**10.62**	**-8.20**

INTERPRETATION

MODEL	R^2	MEANING
4.1 (Manager)	**41.0%**	CEO FE explains 41% of variance in speech that includes CEO
4.1.3 (NonCEO)	**16.3%**	**CEO FE explains only 16% of variance in speech excluding CEO**

WHAT THIS MEANS

> R^2 drops by 24.7pp: CEO fixed effects lose most predictive power when we exclude CEO speech

> CEO regime doesn't spread: Other executives (CFO, IR) maintain their own communication styles

> No "Tone at the Top": CEO's uncertain language does NOT influence the board

CONCLUSION

CEO does NOT set a "Tone at the Top" for uncertainty.

Each executive maintains independent

4.1.3 Regression Coefficients (Non-CEO Model)

> *Linguistic controls still significant; firm controls become insignificant. CEO's uncertain language does NOT propagate to other executives.*

CONTROL COEFFICIENTS (4.1.3 MAIN)

VARIABLE	COEF.	T-STAT	P-VAL	SIG.
Manager_Pres_Unc	**0.0932**	**9.44**	**0.000**	***
Analyst_QA_Unc	**0.0336**	**5.64**	**0.000**	***
Negative_pct	**0.0494**	**3.82**	**0.000**	***
StockRet	**-0.00004**	**-0.25**	**0.800**	
MarketRet	**-0.0002**	**-0.48**	**0.629**	
EPS_Growth	**-0.0004**	**-0.42**	**0.676**	
SurpDec	**-0.0004**	**-0.42**	**0.674**	

** p<0.1, ** p<0.05, *** p<0.01. HC1 Robust.*

MODEL DIAGNOSTICS

METRIC	VALUE
Observations	**55,985**
R-squared	**0.163**
Adj. R-squared	**0.122**
F-statistic	**10.62**
CEO Fixed Effects	**Included**
Year Fixed Effects	**Included**

KEY FINDINGS

> **Linguistic controls:** Still significant (presentation tone sets expectations)

> Firm controls: All become insignificant ($p > 0.6$)

> **R^2 = 16%:** CEO FE explains minimal variance in non-CEO speech

> Conclusion: CEO's regime does NOT influence board's uncertain language

4.1.3 Regression Results: CEO Regime Effect (Non-CEO Speech)

> *Does the CEO's uncertainty persist when they are NOT speaking? Comparison of "Tone at the Top" effect across industries.*

REGRESSION TABLE (DEP VAR: NONCEO_MANAGER_QA_UNCERTAINTY_PCT)

VARIABLE	MAIN	FINANCE	UTILITY
Intercept	1.250*** (0.000)	0.971*** (0.000)	0.651*** (0.001)
Manager_Pres_Uncertainty	0.093*** (0.000)	0.085*** (0.000)	0.183*** (0.000)
Analyst_QA_Uncertainty	0.034*** (0.000)	0.058*** (0.000)	0.070*** (0.001)
Entire_All_Negative	0.049*** (0.000)	0.043** (0.028)	0.018 (0.688)
StockRet	-0.000 (0.800)	0.000 (0.621)	0.000 (0.677)
MarketRet	-0.000 (0.629)	-0.001 (0.213)	0.002 (0.354)
EPS_Growth	-0.000 (0.676)	0.000 (0.997)	-0.004 (0.335)
SurpDec	-0.000 (0.674)	0.001 (0.464)	-0.002 (0.488)
CEO Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Observations	55,985	13,072	2,921
R-squared	0.163	0.158	0.136
Adj. R-squared	0.122	0.118	0.086
F-statistic	10.62	8.04	4.95

*** p<0.01, ** p<0.05, * p<0.1; Standard errors are heteroscedasticity robust (HC1).

4.1.4 EstimateCeoTone.py - Net Sentiment as Personal Style

> *Estimates CEO "Tone" (Net Sentiment = Positive% - Negative%) as a persistent communication trait. Runs 3 models: ToneAll, ToneCEO, ToneRegime.*

MODEL FORMULA

$$\mathbf{DV_NetTone_{it}} = \alpha + \gamma_i + \delta_t + \beta_1 \cdot \text{Pres_NetTone} + \beta_2 \cdot \text{Analyst_QA_NetTone} + \beta_3 \cdot \text{Uncertainty} + \beta_4 \cdot \text{StockRet} + \beta_5 \cdot \text{MarketRet} + \beta_6 \cdot \text{EPS_Growth} + \beta_7 \cdot \text{SurpDec} + \varepsilon_{it}$$

VARIABLE DEFINITIONS

SYMBOL	VARIABLE	DEFINITION
NetTone	Positive% - Negative%	Net sentiment score
γ_i	CEO FE	CEO fixed effect
δ_t	Year FE	Year fixed effect
β_1	Pres_NetTone	Presentation net tone
β_2	Analyst_QA_NetTone	Analyst question net tone
β_3	Uncertainty_pct	Overall uncertainty %
$\beta_{4\text{-}7}$	Firm Controls	StockRet, MarketRet, EPS_Growth, SurpDec

THREE MODELS ESTIMATED

MODEL	DV	WHO'S SPEAKING
ToneAll	Manager_QA_NetTone	CEO + CFO + IR + Others
ToneCEO	CEO_QA_NetTone	CEO only
ToneRegime	**NonCEO_Manager_NetTone**	**CFO + IR + Others (NO CEO)**

RESEARCH QUESTION

Does the CEO's TONE influence other executives?

Same Tone-at-the-Top test as 4.1.3, but for sentiment (positive vs negative) instead of uncertainty.

4.1.4 Model 1: ToneAll (All Managers)

> *DV = Manager_QA_NetTone. Predicts net sentiment for ALL managers (CEO + CFO + IR + Others).*

MODEL SUMMARY

METRIC	VALUE
Dependent Variable	**Manager_QA_NetTone**
Who's Speaking	CEO + CFO + IR + Others
Observations	**57,065**
N CEOs	**2,587**
N Firms	**1,559**
R^2	**0.453 (45.3%)**
Adj. R^2	**0.426**
F-statistic	**20.15**

INTERPRETATION

CEO fixed effects explain 45.3% of variance in all managers' net sentiment. High R^2 because CEO speech is included in the DV.

SAMPLE BREAKDOWN

SAMPLE	OBS	CEOS	R^2
Main	**57,065**	**2,587**	**0.453**
Finance	**13,239**	**569**	**0.511**
Utility	**2,955**	**135**	**0.347**

KEY INSIGHT

Baseline model.

R^2 = 45% mixes CEO's own tone with other managers. Not a clean test of CEO influence on others.

4.1.4 Model 2: ToneCEO (CEO Only)

DV = CEO_QA_NetTone. Predicts net sentiment for CEO's speech ONLY.

MODEL SUMMARY

METRIC	VALUE
Dependent Variable	**CEO_QA_NetTone**
Who's Speaking	CEO only
Observations	**42,062**
N CEOs	**2,013**
N Firms	**1,341**
R^2	**0.277 (27.7%)**
Adj. R^2	**0.241**
F-statistic	**16.97**

CONTROL COEFFICIENTS

VARIABLE	COEF.	P-VAL
CEO_Pres_NetTone	**0.110**	**0.000**
Analyst_QA_NetTone	**0.189**	**0.000**
Uncertainty_pct	**-0.260**	**0.000**
StockRet	**0.001**	**0.032**
MarketRet	**0.003**	**0.000**
EPS_Growth	**-0.002**	**0.461**
SurpDec	**0.008**	**0.000**

HC1 Robust SEs. Green = significant ($p < 0.05$)

INTERPRETATION

CEO fixed effects explain 27.7% of variance in CEO's own net sentiment. This captures the CEO's personal tone style.

4.1.4 Model 3: ToneRegime (Non-CEO Only - Placebo)

> *DV = NonCEO_Manager_QA_NetTone. Predicts net sentiment for CFO + IR + Others (NO CEO). This is the "Tone at the Top" test.*

MODEL SUMMARY

METRIC	VALUE
Dependent Variable	**NonCEO_Manager_QA_NetTone**
Who's Speaking	CFO + IR + Others (NO CEO)
Observations	**55,985**
N CEOs	**2,574**
N Firms	**1,552**
R^2	**0.167 (16.7%)**
Adj. R^2	**0.126**
F-statistic	**10.97**

CONTROL COEFFICIENTS

VARIABLE	COEF.	P-VAL
Manager_Pres_NetTone	**0.161**	**0.000**
Analyst_QA_NetTone	**0.160**	**0.000**
Uncertainty_pct	**-0.185**	**0.000**
StockRet	**0.001**	**0.031**
MarketRet	**0.001**	**0.417**
EPS_Growth	**0.004**	**0.002**
SurpDec	**0.009**	**0.000**

HC1 Robust SEs. Green = significant ($p < 0.05$)

INTERPRETATION

CEO fixed effects explain only 16.7% of variance in non-CEO speech. R^2 drops by 28.6pp vs ToneAll!

4.1.4 Summary: Three-Model Comparison

> *Side-by-side comparison of all three Tone models. R^2 drops sharply when excluding CEO speech.*

MODEL FIT COMPARISON (MAIN SAMPLE)

METRIC	TONEALL (ALL MANAGERS)	TONECEO (CEO ONLY)	TONEREGIME (NON-CEO ONLY)
DV	Manager_QA_NetTone	CEO_QA_NetTone	**NonCEO_Manager_NetTone**
Observations	**57,065**	**42,062**	**55,985**
N CEOs	**2,587**	**2,013**	**2,574**
R^2	**0.453**	**0.277**	**0.167**
Adj. R^2	**0.426**	**0.241**	**0.126**
F-statistic	**20.15**	**16.97**	**10.97**
Prob(F)	**0.000**	**0.000**	**0.000**

CONCLUSION

CEO does NOT set a "Tone at the Top" for sentiment.

CEO FE explains 45% of all-manager tone but only 17% when CEO speech is excluded. Other executives maintain independent communication styles.

4.2 LiquidityRegressions.py - OLS and 2SLS

> *Tests whether Clarity Score and Q&A Uncertainty affect market liquidity around earnings calls. Uses CCCL shift_intensity as instrument for endogenous Uncertainty.*

MODEL SPECIFICATION

$$\Delta \text{Liquidity}_{it} = \alpha + \beta_1 \cdot \text{ClarityScore} + \beta_2 \cdot \text{QA_Uncertainty} + \beta_3 \cdot \text{Controls} + \delta_t + \varepsilon_{it}$$

KEY VARIABLES

VARIABLE	DESCRIPTION
Delta_Amihud	DV: Change in Amihud illiquidity
Delta_Corwin_Schultz	DV: Change in bid-ask spread
Clarity Score	CEO FE from 4.1 (ClarityRegime or ClarityCEO)
QA_Uncertainty_pct	Endogenous: Time-varying uncertainty (instrumented)
shift_intensity_sale_ff48	Instrument: FF48 sales-weighted CCCL exposure

ESTIMATION STRATEGY

PHASE	METHOD	PURPOSE
1	First Stage	Validate instrument (F > 10)
2	OLS	Baseline (no instrument)
3	**2SLS (IV)**	Uncertainty instrumented

CONTROL VARIABLES

> Linguistic: Analyst_QA_Uncertainty_pct, Entire_All_Negative_pct

> Firm: Size, BM, Lev, ROA, CurrentRatio, RD_Intensity, Volatility

> Market: StockRet, MarketRet, EPS_Growth, SurpDec

> Fixed Effects: Year FE

> *Validation of Shift Intensity (CCCL Exposure) as an instrument for Uncertainty. Instrument is strong (F > 10) for both Manager and CEO uncertainty.*

REGRESSION FORMULA: FIRST STAGE

$$\text{QA_Uncertainty}_{it} = \alpha + \beta_1 \cdot \text{CCCL_Exposure}_{it} + \beta_2 \cdot \text{Controls} + \delta_t + \varepsilon_{it}$$

Instrument: CCCL_Exposure = 48-Ind sales-weighted climate change event frequency

REGRESSION TABLE: FIRST STAGE (DEP VAR: QA_UNCERTAINTY_PCT)

VARIABLE	MANAGER Q&A UNC	CEO Q&A UNC
Instrument (Shift Intensity FF48)	-0.554*** (t = -15.53)	-0.449*** (t = -6.49)
Analyst_QA_Unc	0.046*** (p < 0.001)	0.047*** (p < 0.001)
Negative_pct	0.140*** (p < 0.001)	0.149*** (p < 0.001)
Size	-0.013*** (p < 0.001)	-0.006*** (p < 0.001)
BM	0.030*** (p < 0.001)	0.035*** (p < 0.001)
Firm Controls	Yes	Yes
CEO Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes
Observations	56,592	42,469
R-squared	0.054	0.041
F-statistic	241.06	42.10

4.2 Phase 2: OLS Results (No Instrument)

> *Does Clarity or Uncertainty predict liquidity changes? Result: Clarity is associated with wider bid-ask spreads (p<0.05). Uncertainty is insignificant.*

REGRESSION FORMULA: OLS MODEL

$$\text{Likelihood_Change}_{it} = \alpha + \beta_1 \cdot \text{Clarity}_i + \beta_2 \cdot \text{QA_Uncertainty}_{it} + \text{Controls} + \delta_t + \varepsilon_{it}$$

REGRESSION TABLE: OLS MODELS

VARIABLE	DEPENDENT VAR: Δ AMIHUD		DEPENDENT VAR: Δ SPREAD (CORWIN-SCHULTZ)	
	REGIME MODEL	CEO MODEL	REGIME MODEL	CEO MODEL
Clarity Score (Exogenous FE)	0.011 (p = 0.120)	0.004 (p = 0.794)	0.0006** (p = 0.020)	0.0008*** (p = 0.007)
Q&A Uncertainty	-0.016 (p = 0.684)	-0.065 (p = 0.392)	-0.0003 (p = 0.652)	0.0009 (p = 0.214)
Size	-0.023 (p = 0.448)	-0.031 (p = 0.461)	-0.007*** (p < 0.001)	-0.007*** (p < 0.001)
MarketRet	-0.010** (p = 0.011)	-0.013** (p = 0.011)	-0.0002*** (p < 0.001)	-0.0001*** (p = 0.005)
Firm Controls	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Observations	55,564	40,887	55,646	40,946
R-squared	0.001	0.001	0.076	0.072

4.2 Phase 3: 2SLS Results (IV for Uncertainty)

> *Accounting for endogeneity via Instrument (Shift Intensity). Result: Significance disappears. No causal link found between Clarity/Uncertainty and Liquidity.*

REGRESSION FORMULA: 2SLS (INSTRUMENTED)

$$\text{Likelihood_Change}_{it} = \alpha + \beta_1 \cdot \text{Clarity}_i + \beta_2 \cdot \text{Likelihood_Uncertainty}\hat{}_{it} + \text{Controls} + \varepsilon_{it}$$

First Stage: Uncertainty predicted by Shift Intensity (CCCL Exposure).

REGRESSION TABLE: IV MODELS (INSTRUMENTED UNCERTAINTY)

VARIABLE	DEPENDENT VAR: Δ AMIHUD		DEPENDENT VAR: Δ SPREAD (CORWIN-SCHULTZ)	
	REGIME MODEL	CEO MODEL	REGIME MODEL	CEO MODEL
Clarity Score (Exogenous FE)	-1.079 (p = 0.504)	5.729 (p = 0.733)	-0.026 (p = 0.250)	-0.030 (p = 0.730)
QA Uncertainty̑ (Instrumented)	-5.703 (p = 0.500)	25.70 (p = 0.734)	0.011 (p = 0.254)	-0.138 (p = 0.725)
Controls	Yes	Yes	Yes	Yes
Observations	51,053	37,777	51,126	37,830
KP F-Stat (Instrument Strength)	1,256	704	1,257	705

KP F-Statistic > 10 indicates strong instrument but Second Stage results are insignificant.
Large standard errors (and insignificant p-values) suggest OLS results may have been driven by endogeneity.

4.3 Phase 4: Takeover Hazard Methodology

> *Does CEO Language predict the likelihood of receiving a takeover bid? Method: Survival Analysis (Cox PH & Fine-Gray Competing Risks).*

HAZARD FUNCTION SPECIFICATION

$$h(t \mid X) = h_0(t) \cdot \exp(\ \beta_1 \cdot \mathbf{Clarity_i} + \beta_2 \cdot \mathbf{QA_Uncertainty_{it}} + \text{Controls}\ )$$

Event: Receipt of Takeover Bid within 365 days of Earnings Call.
Models: Cox PH (Any Bid) vs. Fine-Gray (Friendly vs. Uninvited).

VARIABLES

VARIABLE	DESCRIPTION
Clarity Score	CEO Fixed Effect (Regime vs. CEO-Specific)
QA Uncertainty	Call-level vagueness (% uncertain words)
Firm Controls	Size, BM, Leverage, ROA (Lagged)

SAMPLE & EVENTS

METRIC	COUNT
Total Calls (Main)	**88,205**
All Takeover Bids	**1,088**
— Friendly / Neutral	**919**
— Hostile / Unsolicited	**169**

4.3 Results: Cox Proportional Hazards (All Bids)

> *ClarityCEO significant (p=0.009, HR=1.14) — higher clarity → higher takeover hazard. ClarityRegime NOT significant.*

COX PH HAZARD RATIOS (ALL TAKEOVERS)

MODEL	VARIABLE	COEF.	HR	P-VALUE	N	EVENTS	C-INDEX
Regime	ClarityRegime	0.025	1.025	0.591	66,597	780	0.591
	QA_Uncertainty	-0.260	0.771	0.062			
CEO	ClarityCEO	0.132	1.142	0.009	51,297	590	0.592
	QA_Uncertainty	-0.061	0.941	0.684			

SIGNIFICANT CONTROLS (REGIME MODEL)

VARIABLE	HR	P-VAL
Size	1.10	0.000
BM	1.30	0.000
Lev	1.87	0.000

INTERPRETATION

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ClarityCEO: HR=1.14 (p=0.009)
1 SD increase in CEO clarity → 14% higher takeover hazard.

Clear CEOs attract acquirers (or their firms are better targets).
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4.3 Fine-Gray Results: Uninvited Takeovers

> *For hostile/unsolicited bids, QA Uncertainty is significant — higher uncertainty → higher uninvited hazard.*

FINE-GRAY SUB-HAZARD RATIOS (UNINVITED: EVENTS=129 REGIME / 94 CEO)

MODEL	VARIABLE	COEF.	SHR	P-VALUE	N	C-INDEX
Regime	ClarityRegime	0.032	1.033	0.765	66,597	0.668
	QA_Uncertainty	0.571	1.770	0.055		
CEO	ClarityCEO	0.150	1.162	0.216	51,297	0.689
	QA_Uncertainty	0.688	1.989	0.034		

SIGNIFICANT CONTROLS

VARIABLE	SHR	P-VAL
Size (Regime)	1.18	0.003
BM (Regime)	1.65	0.000
Lev (CEO)	2.35	0.003

INTERPRETATION

QA Uncertainty: SHR=1.77-1.99 ($p \leq 0.05$)
Vague communication → ~77-99% higher uninvited takeover hazard.

4.3 Fine-Gray Results: Friendly Takeovers

> *For friendly bids, ClarityCEO significant (p=0.02), QA Uncertainty significant (p=0.003) with OPPOSITE sign.*

FINE-GRAY SUB-HAZARD RATIOS (FRIENDLY: EVENTS=651 REGIME / 496 CEO)

MODEL	VARIABLE	COEF.	SHR	P-VALUE	N	C-INDEX
Regime	ClarityRegime	0.020	1.020	0.697	66,597	0.593
	QA_Uncertainty	-0.456	0.634	0.003		
CEO	ClarityCEO	0.126	1.134	0.025	51,297	0.592
	QA_Uncertainty	-0.230	0.795	0.170		

SIGNIFICANT CONTROLS (REGIME MODEL)

VARIABLE	SHR	P-VAL
Size	1.09	0.001
BM	1.20	0.022
Lev	1.88	0.000

CONCLUSION

High uncertainty → more hostile bids, fewer friendly deals.
ClarityCEO → higher friendly takeover hazard.